

Reporting and discussing the results

G0B75A Meta-analysis
Master of statistics

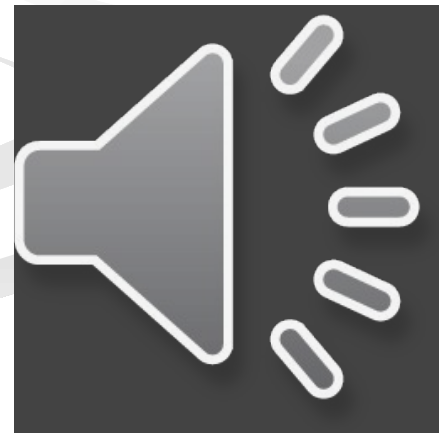
Prof. Wim Van den Noortgate
Faculty of Psychology and Educational Sciences
KU Leuven - University of Leuven, Belgium

Wim.VandenNoortgate@kuleuven.be



Reporting and discussing the results

- a. describing the process
- b. describing the data
- c. describing the meta-analytic results
- d. interpreting the meta-analytic results
- e. checking assumptions
- f. sensitivity analyses



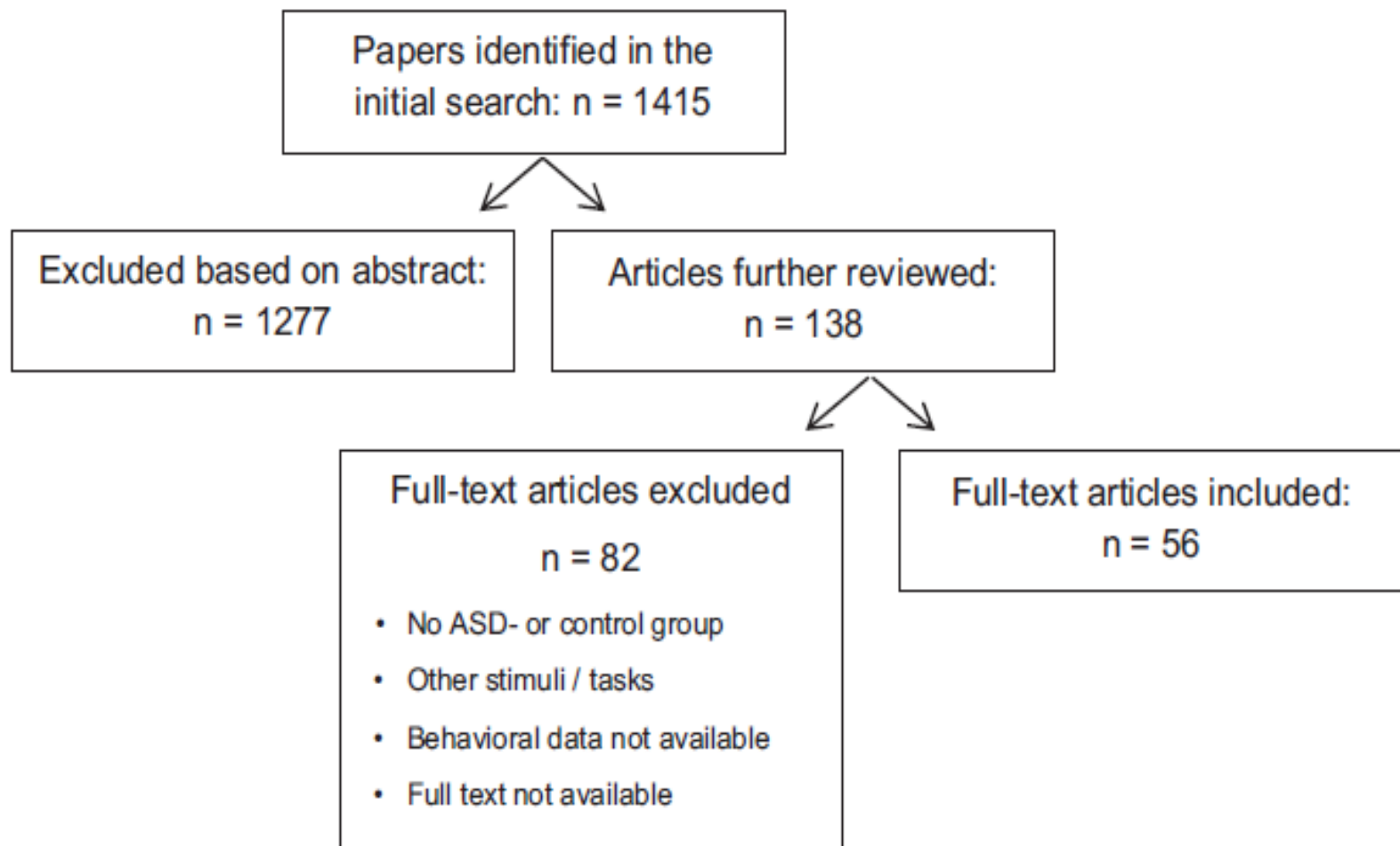
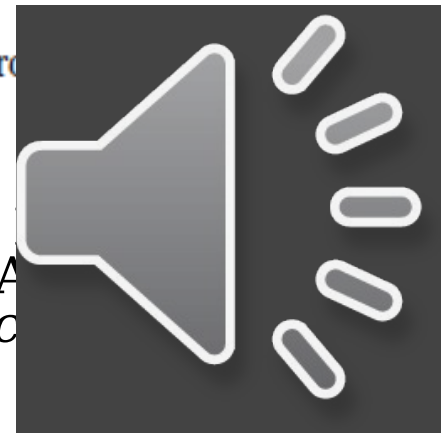
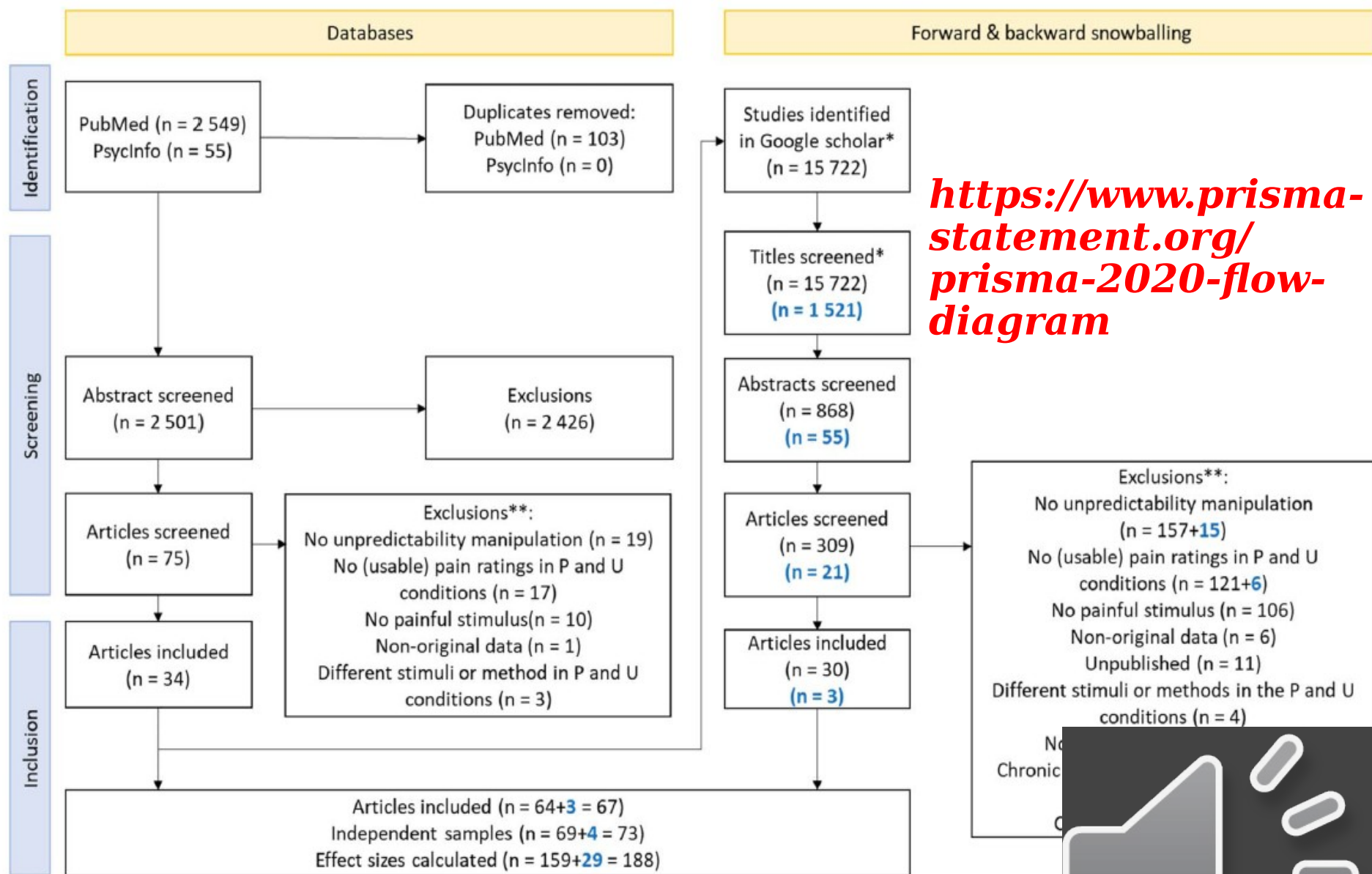


Figure 1. Selection process. This flowchart displays the entire in- and exclusion process to be included in the meta-analysis.

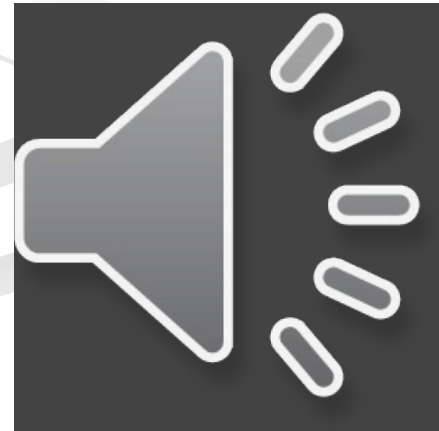
Van der Hallen, R., Evers, K., Brewaeys, K., Van den Wagemans, J. (2015). Global processing takes time: A test on local-global visual processing in ASD. *Psychological Science* (3), 549-573.





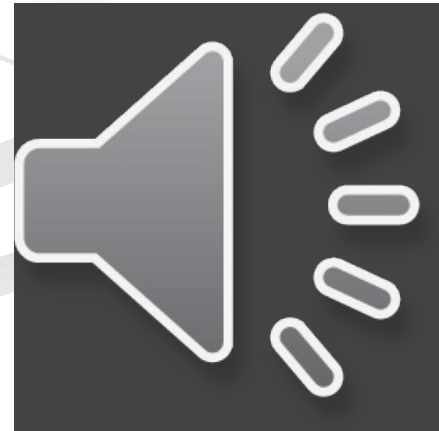
Pavy, F., Zaman, J., Van den Noortgate, W., Scarpa, A., von Leu, D.M. (2024). The effect of unpredictability on the perceived pain in a systematic review and meta-analysis. *Pain*, 165 (8), 1702-1718.

- Effect size calculation,
- software used,
- Decisions made, and reasons why,
- Problems encountered,
- ...



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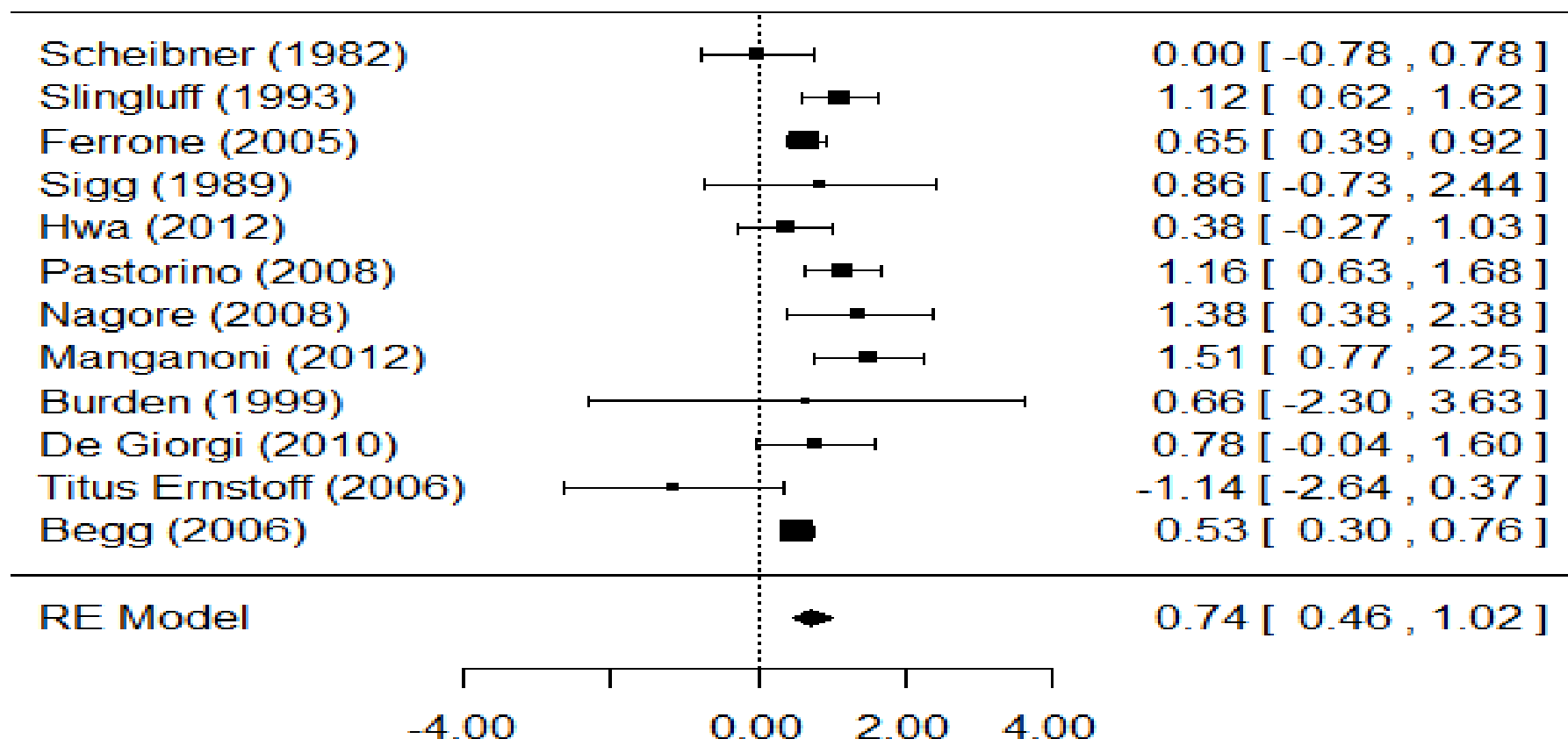


Appendix B. Overview of study characteristics and effect sizes for all study samples

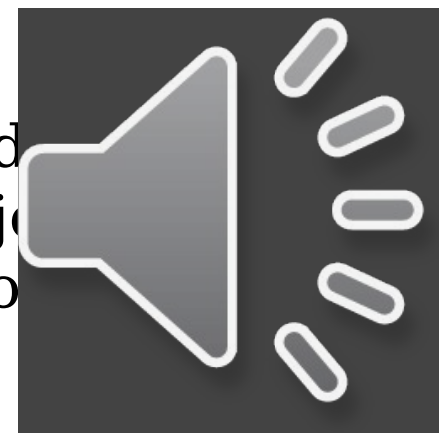
Publication characteristics		Leamer characteristics			Research design features						Video materials				Test features			Effect size statistics	
Study	Publication type (study ID)	L1	Prof	Instr level	Aim	Des	L2	N con group	N treat group	Pilot test	Aut video	Video con	Nº view	Length	Prof det	Test lc	Test voc	g lc	g voc
Markham (1989)	1 (0001)	1	1–3	1	1	1	1	76	76	1	2	2	1	7'	2	1		0.541	
Danan (1992)	1 (0002)	2	1	1	2	2	3	7	5	1	3	3	3	5'	3		2		1.038
Neuman and Koskinen (1992)	1 (0003)	1	1–3	3	3	2	1	37	32	0	2	2	4	±58'	1	2	3	0.195	0.335
Baltova (1999b)	2 (0004)	2	1	3	3	2	3	30	34	1	1	1	3	7'30"	3	2	2	1.497(***)	1.538(***)
Chung (1999)	1 (0005)	1	1	2	1	2	1	170	170	0	3	3	2	±12'	3	1		0.682(**)	
Huang and Eskey (1999–2000)	1 (0006)	5	2	4	3	1	1	15	15	0	3	3	2	14'	1	1	2	1.293(***)	1.007(**)
Markham (1999)	1 (0007)	1	3	1	2	1	1	57	61	1	2	2	1	25'	1		1		0.870(***)
Liversidge (2000)	2 (0008)	1	1, 2	1	3	2	1	69	79	1	1	4	2	22'	4	1	2	0.773(***)	0.732(**)
Markham (2001)	1 (0009)	1	3	2	1	2	1	54	54	0	1	1	1	19'36"	1	2		0.711(*)	
Markham et al. (2001)	1 (0010)	2	2	1	1	2	2	68	43	1	4	1	1	7'	1	3		0.548(**)	
Hwang (2003)	2 (0011)	1	2	2	3	1	1	40	40	1	3	3	4	15'	1	1	1	1.646(***)	1.214(***)
Markham and Peter (2003)	1 (0012)	2	2	1	1	2	2	63	85	1	4	1	1	7'	1	1		0.967(***)	
Hernandez (2004)	2 (0013)	3	2	2	3	2	1	30	32	1	1	4	2	8'05"	1				
Park (2004)	2 (0014)	1	1–3	1	1	2	1	89	89	0	1	1	2	±12'	1				
Taylor (2005)	1 (0015)	2	1	1	1	1	2	36	35	1	3	3	1	10'	1				

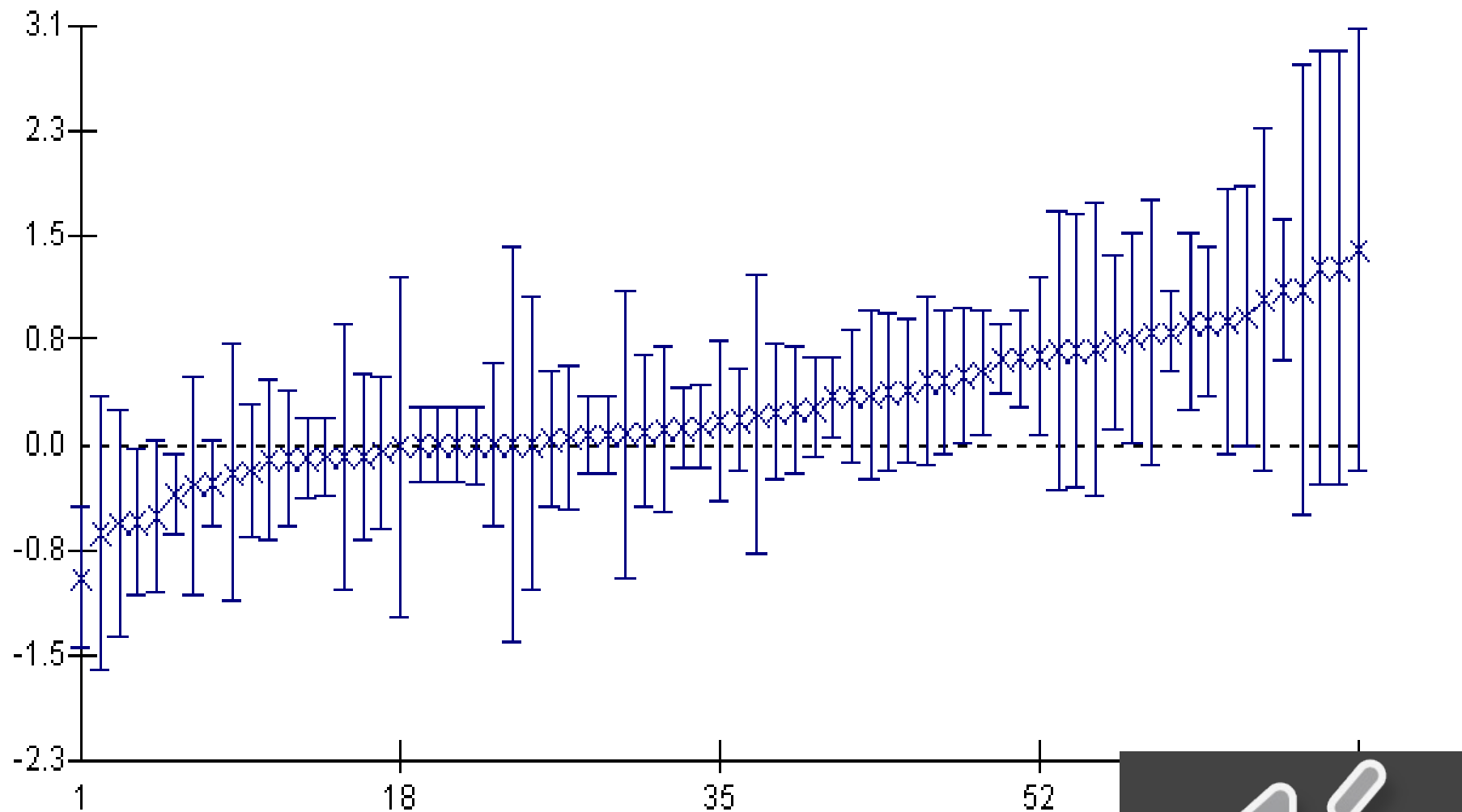
Montero Perez, M., Van den Noortgate, W., De
(2013). Captioned video for L2 listening and v
learning: A meta-analysis. *System*, 41 (3), 720



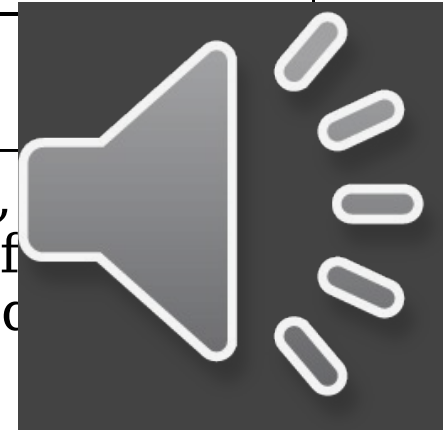


Marasigan, V., Schmitt, E., Garmyn, M., Van der
 Noortgate, W. (2014). A meta-analysis on major
 risk factors of multiple primary cutaneous melanoma.
Dermatology, 2 (2), 84-92.



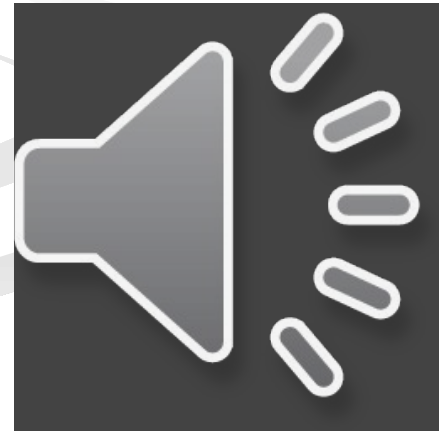


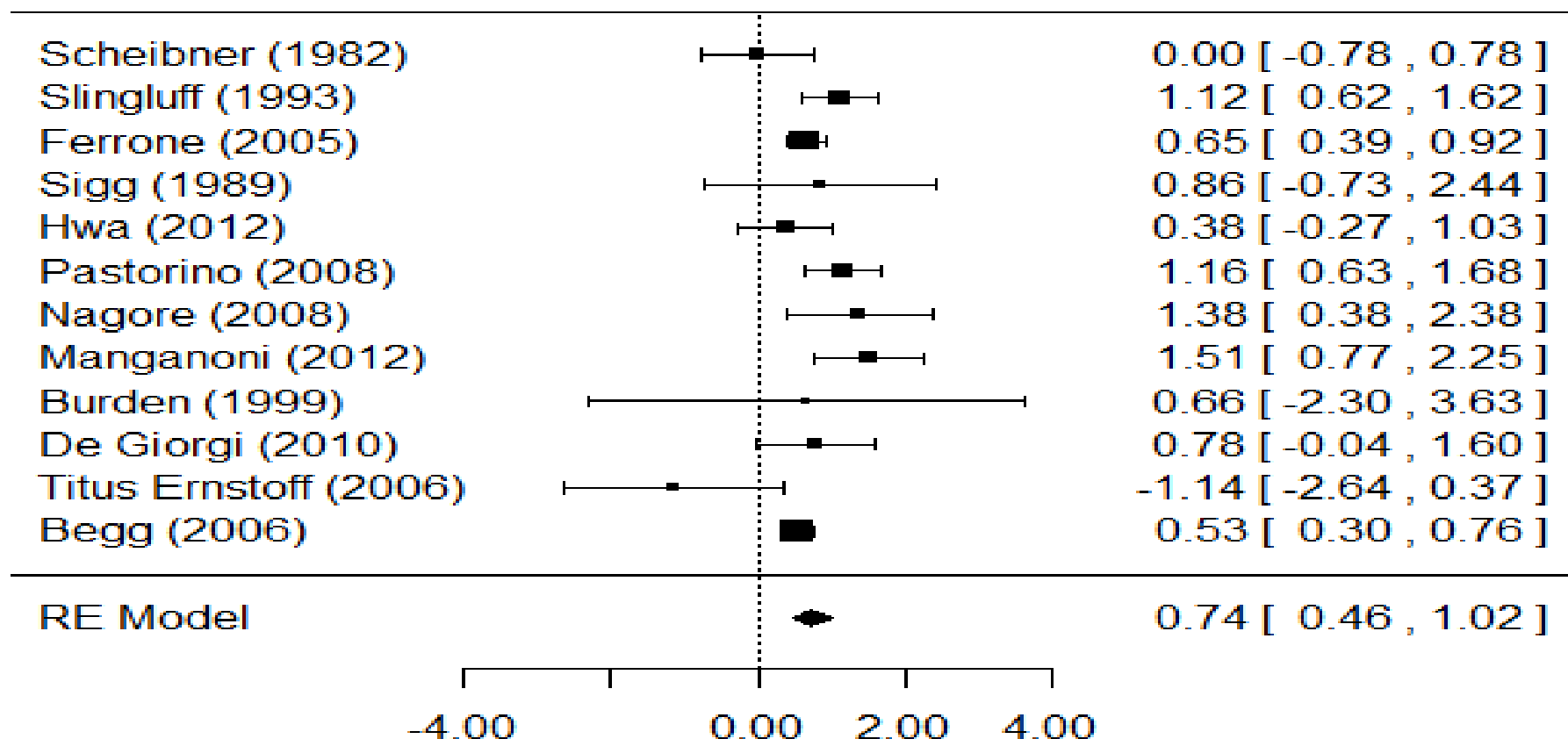
Geeraert, L., Van den Noortgate, W., Grietens, H., (2004). The effects of early prevention programs for young children at risk for physical child abuse and neglect: a meta-analysis. *Child Maltreatment*, 9, 277-291.



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Marasigan, V., Schmitt, E., Garmyn, M., Van der Meulen, W., & Noortgate, W. (2014). A meta-analysis on major risk factors of multiple primary cutaneous melanoma. *Journal of the American Academy of Dermatology*, 2 (2), 84-92.

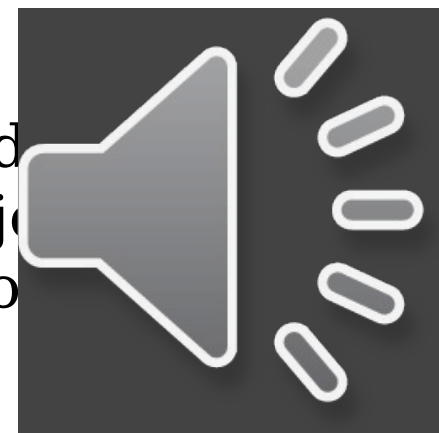


Table 3

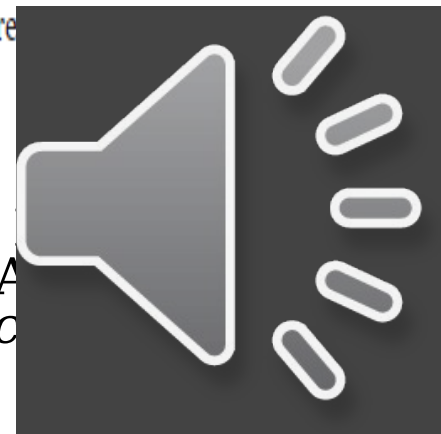
The Level of Visual Processing as a Moderator Variable

	Number of observed ES	<i>df</i>	<i>F</i>	<i>p</i> > <i>F</i>	<i>g</i>	<i>df</i>	<i>t</i>	<i>p</i> > <i>t</i>
Primary (task-relevant) level of processing		(1, 233)	5.90	.0159*				
Local	193				−0.01	40.8	−0.06	.9487
Global	87				−0.47	74.5	−2.03	.0462*
Primary (task-relevant) and secondary (task-irrelevant) level(s) of processing		(3, 230)	2.28	.0797				
Local–Local	30				−0.18	140.0	−0.60	.5500
Local–Global	163				0.03	46.6	0.15	.8823
Global–Global	16				−0.30	204.0	−0.78	.4350
Global–Local	71				−0.52	83.5	−2.13	.0358*

Note. Whereas “Local–Local” and “Global–Global” refer to instances where the secondary (task-irrelevant) level of visual processing is absent or nonconflicting, “Local–Global” and “Global–Local” refer to instances where the secondary (task-irrelevant) level of visual processing is present and conflicting with the primary level of visual processing (also see Table 1); data displayed include both accuracy and reaction times; *g* refers to Hedges’ *g* effect size.

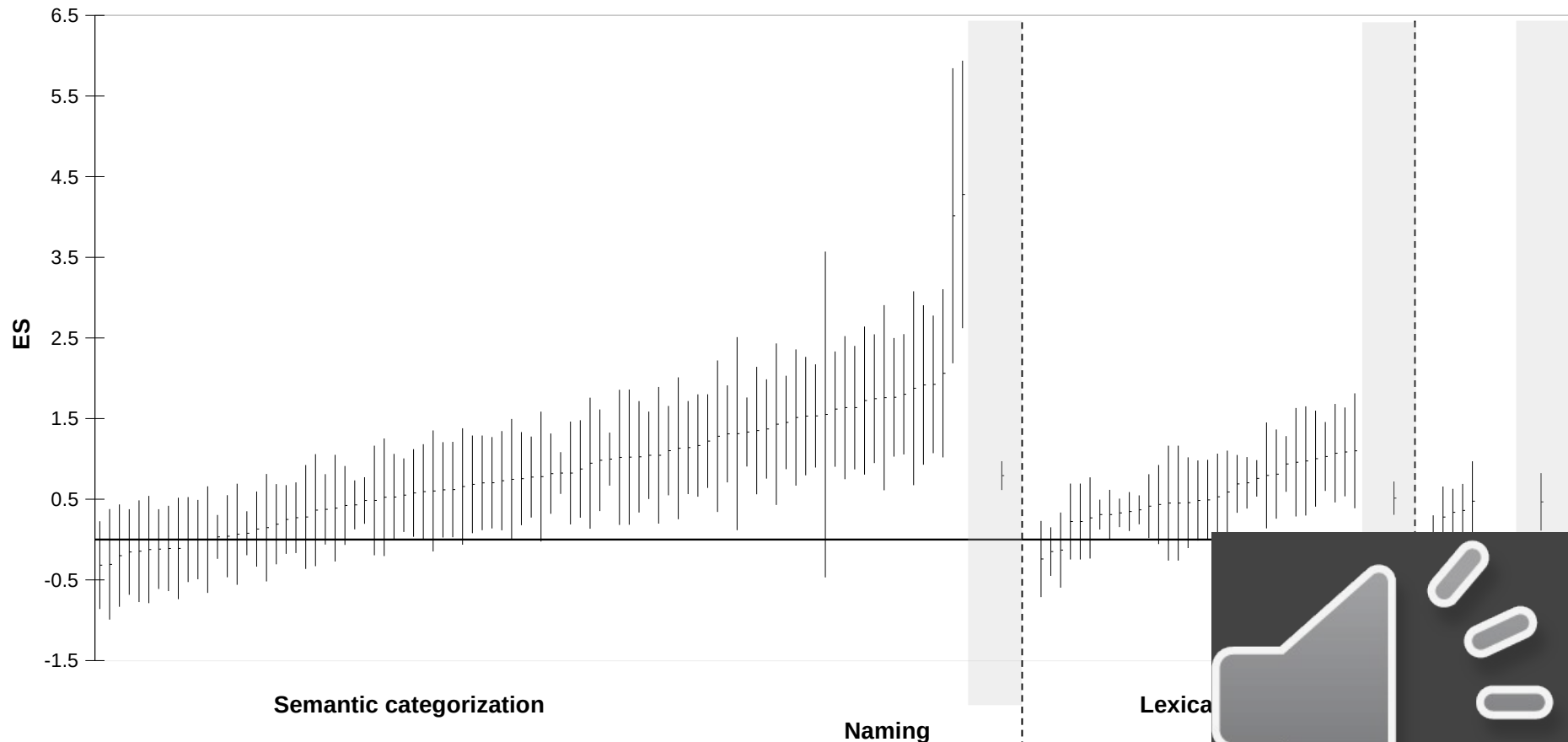
**p* < .05.

Van der Hallen, R., Evers, K., Brewaeys, K., Van den Wagemans, J. (2015). Global processing takes time: A test on local-global visual processing in ASD. *Psychological Science*, 26(3), 549-573.

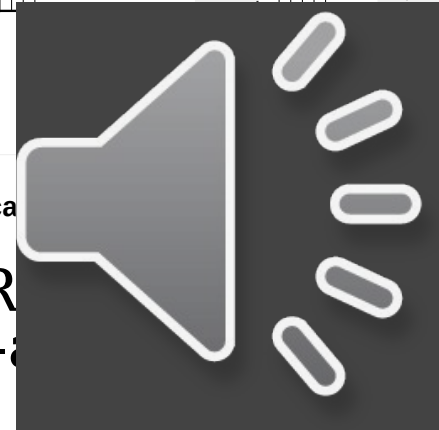


A sorted caterpillar plot

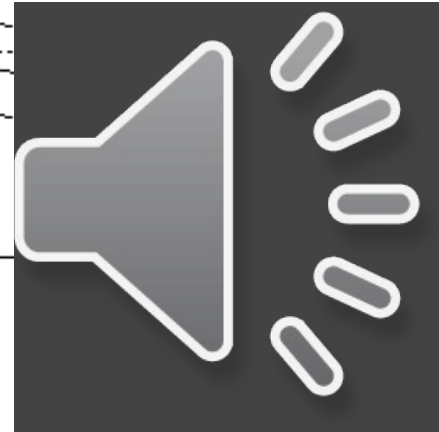
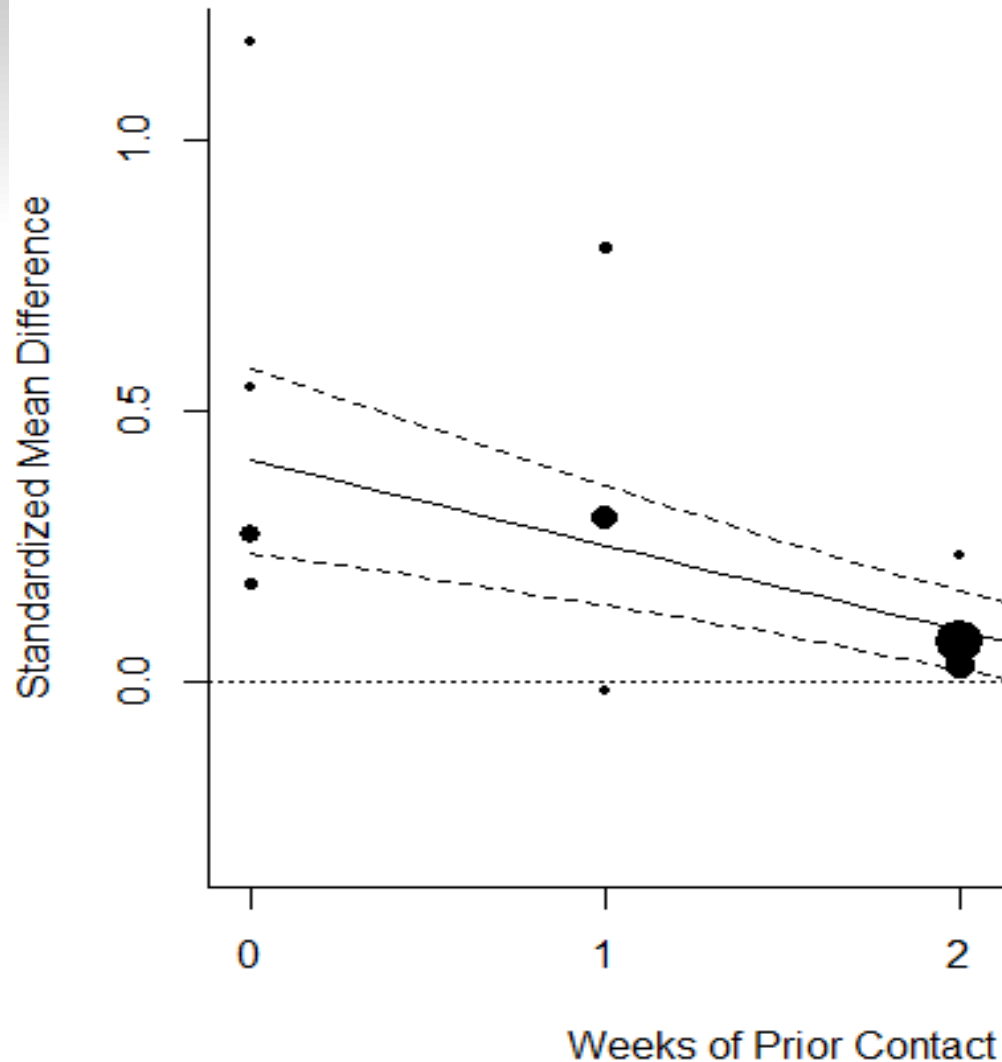
Observed effect sizes for the 3 tasks



Van den Bussche, E., Van den Noortgate, W., & R (2009). Mechanisms of masked priming: A meta-analysis. *Psychological Bulletin*, 135, 452-477.



A Regression plot



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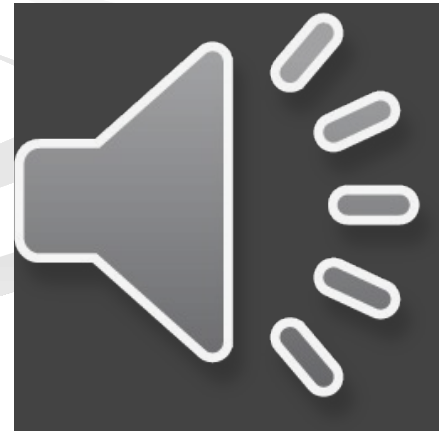


Remark on external validity of the study:
No random sampling of studies!



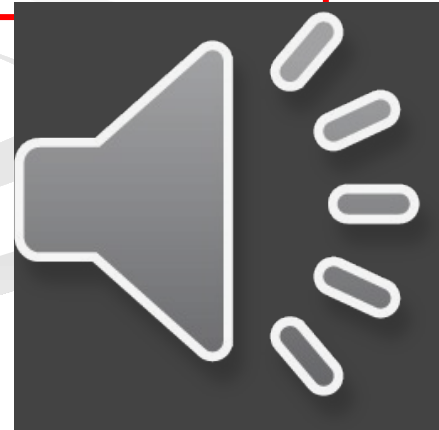
Remark on internal validity of the study: **No random assignment of studies!**

Be aware of confounding
moderators!

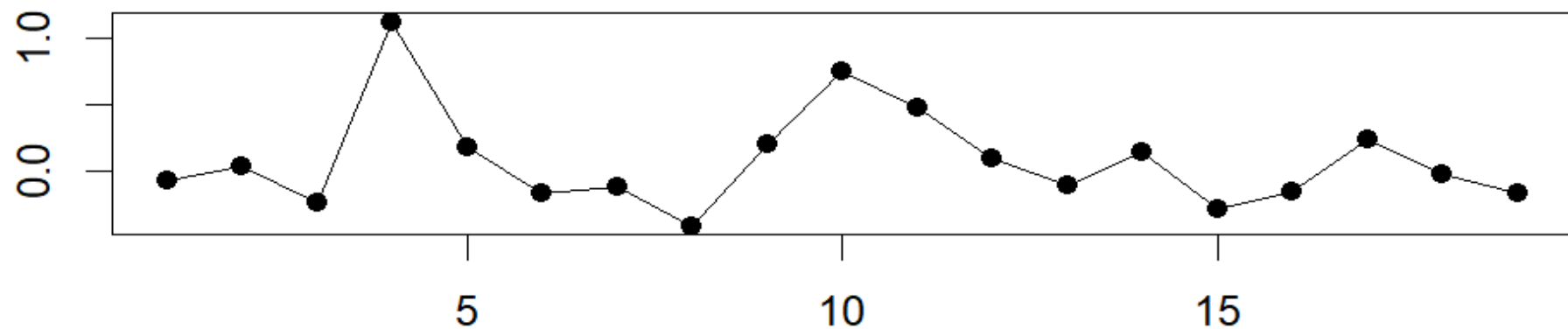


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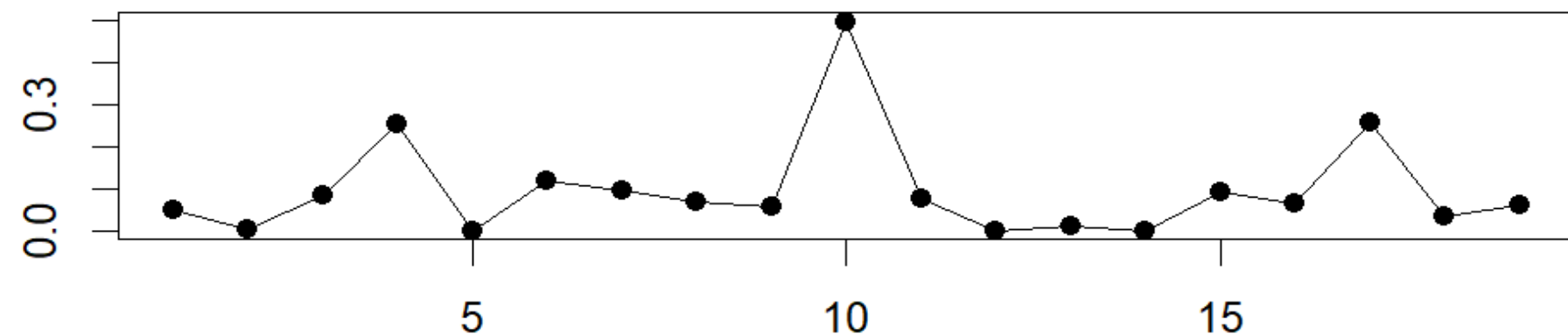
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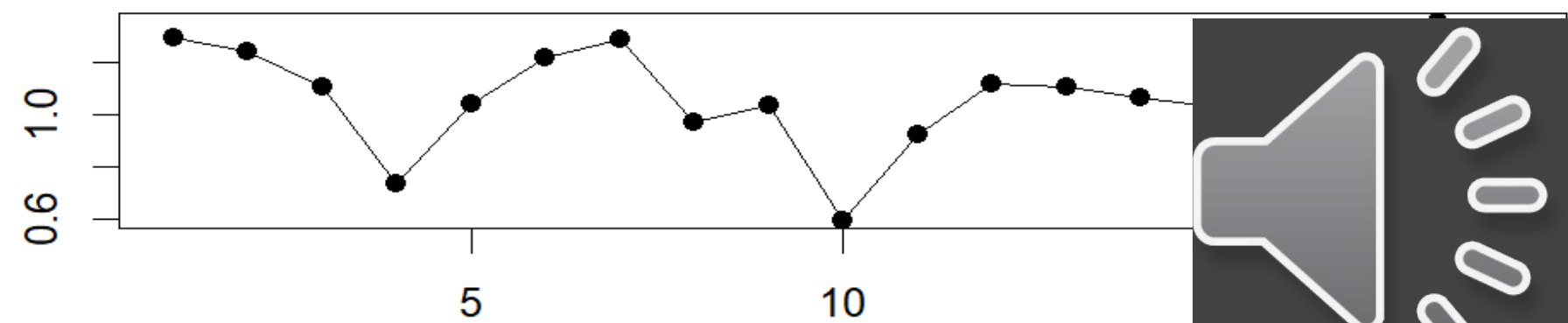
Studentize residuals



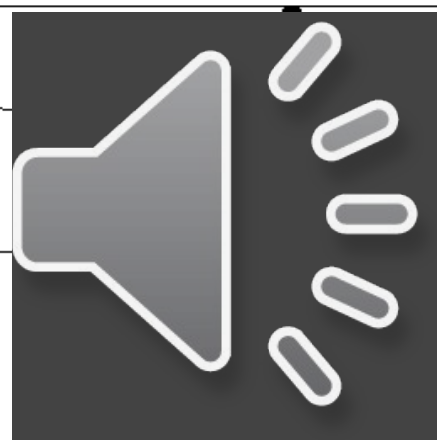
Cook's Distance



Covariance ratio

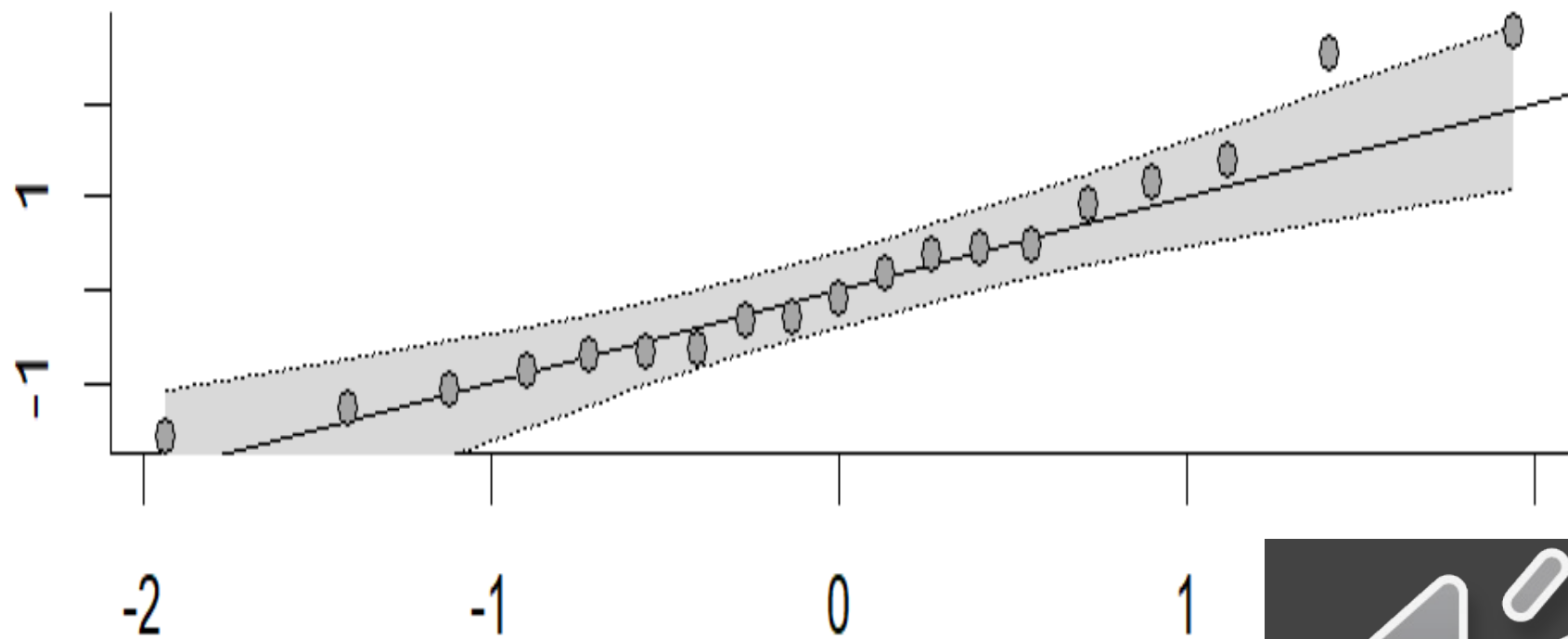


Observed Outcome

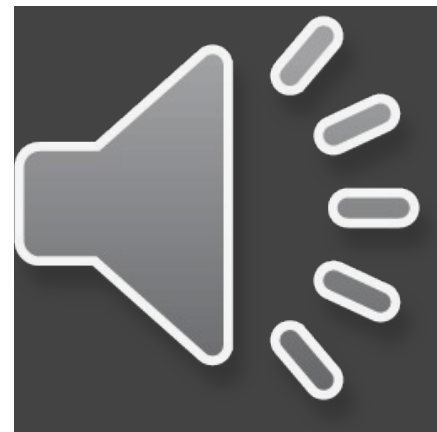


Normal Q-Q Plot

Sample Quantiles



Theoretical Quantiles



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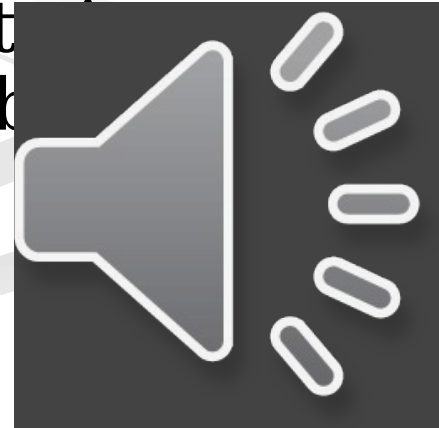


What is the effect of your decisions?



Sensitivity analysis: how robust are conclusions?

- **outliers:** conduct analysis with and without outliers
- **calculation effect sizes :** several analyses
- **publication bias:** analysis with and without unpublished results
- **design & quality:** compare results from studies with strong design or good quality, with those of all studies
- **researcher:** literature search, effect calculation, coding quality, ..., done by researchers
- meta-analytic **model** used
- ...



**GOOD REPORTING IS NOT AN OPTIONAL EXTRA.
IT IS AN ESSENTIAL COMPONENT OF DOING GOOD
RESEARCH.**

